

Propositional Logic

COMP2711 Discrete Mathematical Tools

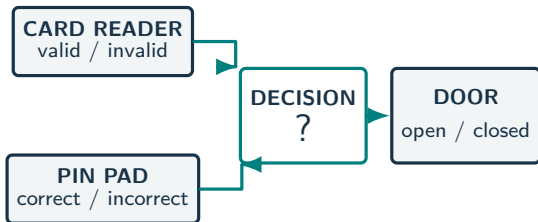
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SECTION 1

Motivation

A one-line security bug

Your first task on a security team: program a laboratory door.

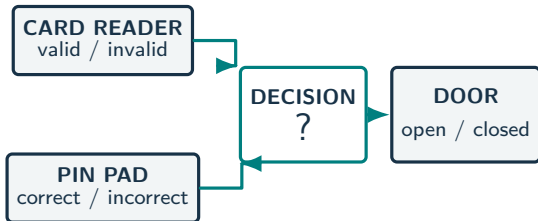


A one-line security bug

Your first task on a security team: program a laboratory door.

A rushed implementation

```
if card OR pin: open
```



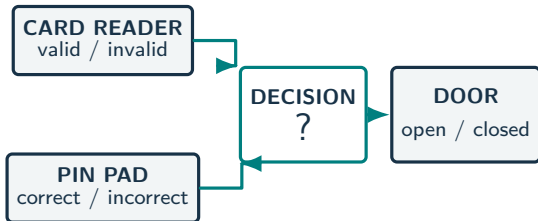
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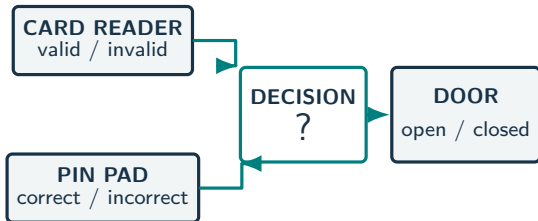
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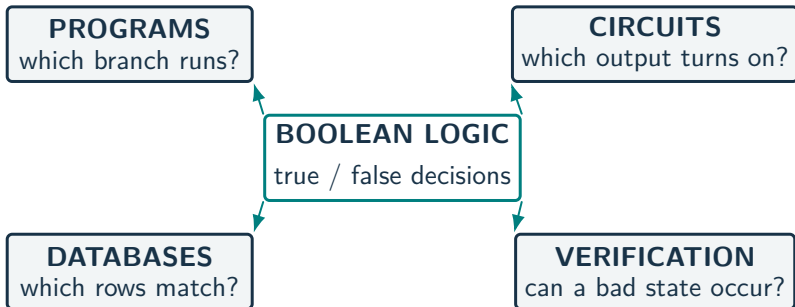
Correct decision

```
if card AND pin: open
```

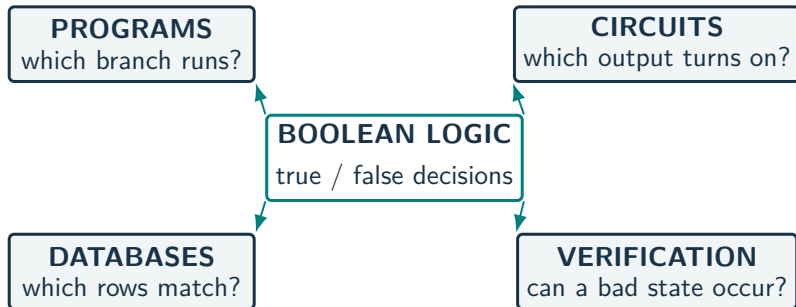


One logical word changes which states the program accepts.

Logic in computer science

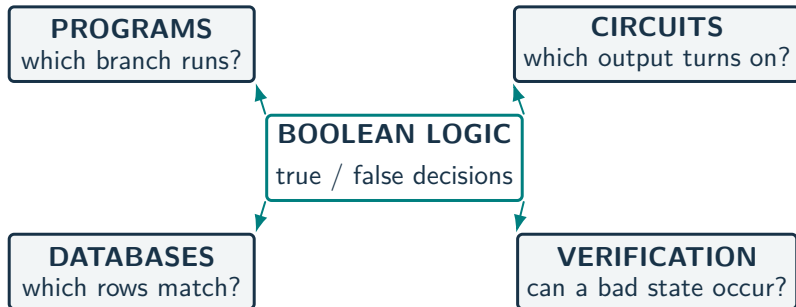


Logic in computer science



A requirement becomes a formula; the formula becomes a machine decision.

Logic in computer science



A requirement becomes a formula; the formula becomes a machine decision.

Logic lets us ask: **Does the decision work for every possible input?**

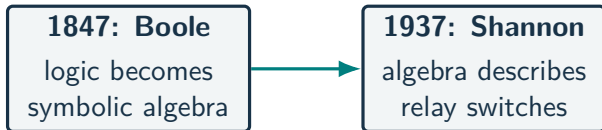
Boole and Shannon

1847: Boole

logic becomes
symbolic algebra

Sources: Library of Congress; MIT.

Boole and Shannon

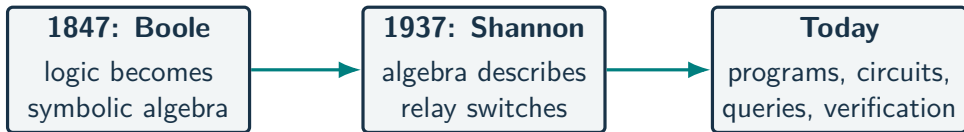


As an MIT master's student, Shannon recognized that a relay switch behaves like a truth value:

TRUE / FALSE $\longleftrightarrow$ **ON / OFF**

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Boole and Shannon



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TRUE / FALSE $\longleftrightarrow$ **ON / OFF**

A machine could now evaluate a logical formula.

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SECTION 2

Formulas and truth tables

Propositions

MUST UNDERSTAND

Proposition. A declarative sentence with exactly one truth value.

Sentence	Proposition?
HKUST is in Hong Kong.	
17 is divisible by 3.	
$x + 2 = 7$.	
Submit the assignment.	

Propositions

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Proposition. A declarative sentence with exactly one truth value.

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HKUST is in Hong Kong.	yes
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Submit the assignment.	no

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17 is divisible by 3.	yes
$x + 2 = 7$.	not yet
Submit the assignment.	no

False statements can be propositions. Their truth value must be definite.

Atomic propositions

MUST UNDERSTAND

Atomic proposition. One complete claim treated as a single unit.

For the door example, use readable names:

Name	Complete claim
<i>card</i>	the card is valid
<i>pin</i>	the PIN is correct
<i>open</i>	the door opens

Each atom names a complete claim. It is not an object or number.

Formulas

MUST UNDERSTAND

Logical formula. An atomic proposition, or propositions combined by logical connectives.

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Arithmetic formula	Logical formula
$(x + y) - z$	$(p \vee q) \wedge \neg r$
x, y, z are numbers	p, q, r are T or F
rules: $+, -, \times, \div$	connectives: $\neg, \wedge, \vee, \oplus$

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x, y, z are numbers	p, q, r are T or F
rules: $+, -, \times, \div$	connectives: $\neg, \wedge, \vee, \oplus$

Same pattern: **variables** combined by **rules**.

Connectives

MUST UNDERSTAND

Connective. An operation that combines one or more formulas to form a new formula.

Name	Formula	Read as	True when
negation	$\neg p$	not p	p is false
conjunction	$p \wedge q$	p and q	both are true
disjunction	$p \vee q$	p or q	at least one is true
exclusive disjunction	$p \oplus q$	p XOR q	exactly one is true

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exclusive disjunction	$p \oplus q$	p XOR q	exactly one is true

OR: at least one is true.

XOR: exactly one is true.

Formula structure

Consider

$$(p \vee q) \wedge \neg p$$

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1. Compute the left part: $p \vee q$.

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1. Compute the left part: $p \vee q$.
2. Compute the right part: $\neg p$.

Formula structure

Consider

$$(p \vee q) \wedge \neg p$$

1. Compute the left part: $p \vee q$.
2. Compute the right part: $\neg p$.
3. Combine them with the outer $\wedge$.

The outer $\wedge$ is the **main connective**: it is evaluated last.

Truth assignments

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For two atoms:

(T, T), (T, F), (F, T), (F, F).

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With n atoms, there are 2^n assignments.

For two atoms:

(T, T), (T, F), (F, T), (F, F).

A truth table checks every possible case.

Basic truth tables

p	$\neg p$
T	F
F	T

p	q	
T	T	
T	F	
F	T	
F	F	

Basic truth tables

p	$\neg p$
T	F
F	T

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Basic truth tables

p	$\neg p$
T	F
F	T

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
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Basic truth tables

p	$\neg p$
T	F
F	T

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T	T	T	F
T	F	F	T	T
F	T	F	T	T
F	F	F	F	F

$\vee$: at least one. $\oplus$: exactly one.

Truth-table construction

Evaluate $(p \vee q) \wedge \neg p$.

p	q		
T	T		
T	F		
F	T		
F	F		

Truth-table construction

Evaluate $(p \vee q) \wedge \neg p$.

p	q	$p \vee q$	
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T	F	T	F	F
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Evaluate $(p \vee q) \wedge \neg p$.

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
T	T	T	F	F
T	F	T	F	F
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The formula is true exactly when $p = F$ and $q = T$.

Formula classification

MUST UNDERSTAND

Tautology and contradiction. A tautology is always true; a contradiction is always false.

They behave like constant expressions: the output does not depend on the input.

Behavior	Arithmetic example	Logical examples
always 1	$x - x + 1$	tautologies: $p \vee \neg p$, $q \vee \neg q$

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changes	$x + 1$	contingencies: $p \wedge q, p \vee q$

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always 0	$x - x$	
changes	$x + 1$	contingencies: $p \wedge q, p \vee q$

A truth table asks the same question: **does the final value ever change?**

Logical equivalence

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Logical equivalence. $A \equiv B$ means that A and B agree on every truth assignment.

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Arithmetic identity	Logical equivalence
$(x + 1)(x - 1) = x^2 - 1$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
Same number for every x .	Same truth value for every p, q .

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Same number for every x .	Same truth value for every p, q .

We do not choose a particular x —or a particular truth-table row.

De Morgan's law

Show that $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

De Morgan's law

Show that $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

The shaded columns match in every row.
Therefore, the formulas are equivalent.

SECTION 3

Conditionals

Implication

MUST UNDERSTAND

Implication. $p \rightarrow q$ means: whenever p is true, q must be true.

p : I miss the bus. q : I am late for class.

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p	q	$p \rightarrow q$	What happened?
T	T	T	Missed the bus; late. The rule holds.
T	F	F	Missed the bus; not late. The rule fails.
F	T	T	Did not miss the bus; rule not tested.
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Only $p = \text{T}$, $q = \text{F}$ makes the implication false.

Implication

MUST UNDERSTAND

Implication. $p \rightarrow q$ means: whenever p is true, q must be true.

“If the world ended yesterday, then I am an alien.”

The if-part is false, so the implication is not tested.

$$p = \text{F} \quad \implies \quad p \rightarrow q = \text{T}$$

Necessary and sufficient conditions

$$p \rightarrow q$$

Role	Meaning
p is sufficient for q	p guarantees q

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Example:

$$4 \mid n \rightarrow 2 \mid n.$$

$4 \mid n$ is **sufficient** for $2 \mid n$.

$2 \mid n$ is **necessary** for $4 \mid n$.

Converse and contrapositive

Let p mean “It is a dog,” and let q mean “It is an animal.”

Statement	Form	Example	Result
original	$p \rightarrow q$	dog $\rightarrow$ animal	true

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original	$p \rightarrow q$	dog $\rightarrow$ animal	true
converse	$q \rightarrow p$	animal $\rightarrow$ dog	false: cat
inverse	$\neg p \rightarrow \neg q$	not dog $\rightarrow$ not animal	false: cat

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converse	$q \rightarrow p$	animal $\rightarrow$ dog	false: cat
inverse	$\neg p \rightarrow \neg q$	not dog $\rightarrow$ not animal	false: cat
contrapositive	$\neg q \rightarrow \neg p$	not animal $\rightarrow$ not dog	true

$$p \rightarrow q \equiv \neg q \rightarrow \neg p.$$

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p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Read: p if and only if q .

Biconditional

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p	q	$p \leftrightarrow q$
T	T	T
T	F	F
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F	F	T

Read: p if and only if q .

Equivalent to two implications:

$$A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A).$$

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Equivalent to two implications:

$$A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A).$$

Without arrows:

$$A \leftrightarrow B \equiv (A \wedge B) \vee (\neg A \wedge \neg B)$$

Both are true, or both are false.

Biconditional and logical equivalence

Notation	Status	Reading
$A \leftrightarrow B$ $A \equiv B$	formula	In one particular case: do A and B agree?

Is $A \equiv B$ also a formula? After values are assigned, $A \leftrightarrow B$ is T or F.

Biconditional and logical equivalence

Notation	Status	Reading
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$A \equiv B$	judgment	For every assignment, A and B agree.

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No particular variables or assignment need to be named.

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$A \equiv B$ exactly when $A \leftrightarrow B$ is always T.

No particular variables or assignment need to be named.

Truth tables and equivalence laws can prove the judgment.

Translation

Three steps

1. Define the atomic propositions.
2. Find the main logical phrase.
3. Add parentheses, then read the formula back.

Patterns

English	Formula
p if q	
p only if q	
p iff q	
neither p nor q	
not both p and q	

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p only if q	$p \rightarrow q$
p iff q	$p \leftrightarrow q$
neither p nor q	$\neg p \wedge \neg q$
not both p and q	$\neg(p \wedge q)$

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p only if q	$p \rightarrow q$
p iff q	$p \leftrightarrow q$
neither p nor q	$\neg p \wedge \neg q$
not both p and q	$\neg(p \wedge q)$

After “only if” comes the **necessary** condition.

Lab door rule

The door opens exactly when the card is valid and the PIN is correct.

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exactly when $\longrightarrow \leftrightarrow$

card valid and PIN correct $\longrightarrow card \wedge pin$

$open \leftrightarrow (card \wedge pin)$.

Left to right: no unauthorized opening.

Right to left: correct credentials must work.

Lab door verification

<i>card</i>	<i>pin</i>	<i>open</i>	$card \wedge pin$	$open \leftrightarrow (card \wedge pin)$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	F
T	F	F	F	T
F	T	T	F	F
F	T	F	F	T
F	F	T	F	F
F	F	F	F	T

Lab door verification

<i>card</i>	<i>pin</i>	<i>open</i>	$card \wedge pin$	$open \leftrightarrow (card \wedge pin)$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	F
T	F	F	F	T
F	T	T	F	F
F	T	F	F	T
F	F	T	F	F
F	F	F	F	T

Each false row reveals a bug:
unauthorized access or rejected valid credentials.

SECTION 4

Problem solving

Problem

Show that

$$(p \rightarrow q) \wedge (p \rightarrow \neg q) \equiv \neg p.$$

Prove that both sides have the same truth value for every p and q .

Natural-language reasoning

Temporarily, let p mean “the door opens” and q mean “the alarm sounds.”

$$p \rightarrow q$$

If the door opens, the alarm sounds.

$$p \rightarrow \neg q$$

If the door opens, the alarm does not sound.

Natural-language reasoning

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door opens $\implies$ alarm sounds and does not sound **impossible**

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If the door opens, the alarm sounds.

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door opens $\implies$ alarm sounds and does not sound **impossible**

Both rules can be true only when the door stays closed: $p = \text{F}$.

Therefore, the left side is true exactly when $\neg p$ is true.

Two solution methods

Method	What to do
truth table	Check all four assignments and compare the final columns.
equivalence laws	Rewrite the left side step by step until it becomes $\neg p$.

Both methods prove the same claim in different ways.

Truth-table solution

Compare the left side with $\neg p$ on all four assignments.

p	q	$\neg q$	$p \rightarrow q$	$p \rightarrow \neg q$	left side	$\neg p$
T	T	F	T	F	F	F
T	F	T	F	T	F	F
F	T	F	T	T	T	T
F	F	T	T	T	T	T

Truth-table solution

Compare the left side with $\neg p$ on all four assignments.

p	q	$\neg q$	$p \rightarrow q$	$p \rightarrow \neg q$	left side	$\neg p$
T	T	F	T	F	F	F
T	F	T	F	T	F	F
F	T	F	T	T	T	T
F	F	T	T	T	T	T

The shaded columns match on every row. Therefore,

$$(p \rightarrow q) \wedge (p \rightarrow \neg q) \equiv \neg p.$$

Equivalence-law method

Logical arithmetic

Arithmetic	Logic
replace an expression by an equal one same number	replace a formula by an equivalent one same truth value on every assignment

Start with the left side. Every step must preserve its truth value on every assignment.
Reach $\neg p$.

Equivalence-law method

Useful equivalence laws

Purpose	Law
remove arrows	$A \rightarrow B \equiv \neg A \vee B$ $p \rightarrow \neg q \equiv \neg p \vee \neg q$
distribute	$(A \vee B) \wedge (A \vee C) \equiv A \vee (B \wedge C)$
expose a contradiction	$q \wedge \neg q \equiv \text{F}$
remove F	$A \vee \text{F} \equiv A$

Use a law only when its pattern matches the formula.

Equivalence-law method

Distributive law

$$(A \vee B) \wedge (A \vee C) \equiv A \vee (B \wedge C)$$

Match the repeated part:

$$A = \neg p, \quad B = q, \quad C = \neg q.$$

$$(\neg p \vee q) \wedge (\neg p \vee \neg q) \equiv \neg p \vee (q \wedge \neg q).$$

Equivalence-law solution

Start from the left side. Preserve its truth value on every assignment.

$$(p \rightarrow q) \wedge (p \rightarrow \neg q)$$

Equivalence-law solution

Start from the left side. Preserve its truth value on every assignment.

$$\begin{aligned}(p \rightarrow q) \wedge (p \rightarrow \neg q) \\ \equiv (\neg p \vee q) \wedge (\neg p \vee \neg q)\end{aligned}$$

remove arrows

Equivalence-law solution

Start from the left side. Preserve its truth value on every assignment.

$$(p \rightarrow q) \wedge (p \rightarrow \neg q)$$

$$\equiv (\neg p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv \neg p \vee (q \wedge \neg q)$$

remove arrows

distribute

Equivalence-law solution

Start from the left side. Preserve its truth value on every assignment.

$$(p \rightarrow q) \wedge (p \rightarrow \neg q)$$

$$\equiv (\neg p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv \neg p \vee (q \wedge \neg q)$$

$$\equiv \neg p \vee F$$

remove arrows

distribute

contradiction

Equivalence-law solution

Start from the left side. Preserve its truth value on every assignment.

$$\begin{aligned}(p \rightarrow q) \wedge (p \rightarrow \neg q) \\ &\equiv (\neg p \vee q) \wedge (\neg p \vee \neg q) && \text{remove arrows} \\ &\equiv \neg p \vee (q \wedge \neg q) && \text{distribute} \\ &\equiv \neg p \vee F && \text{contradiction} \\ &\equiv \boxed{\neg p} && \text{remove F.}\end{aligned}$$

We reached the right side, so the formulas are equivalent.

Knights and knaves

On an island:

- a **knight** always tells the truth;
- a **knave** always lies.

You meet A and B.

A says

"B is a knight."

B says

"A is a knight."

What are their types?

Modeling the first exchange

p A is a knight
 q B is a knight
 S_A statement made by A
 S_B statement made by B

Speaker rule

$$p \leftrightarrow S_A \quad q \leftrightarrow S_B$$

Type and statement have the same truth value.

What did they say? $S_A = q, \quad S_B = p.$

Substitute into the speaker rule:

$$(p \leftrightarrow q) \wedge (q \leftrightarrow p) \equiv p \leftrightarrow q.$$

Thus p and q agree: both true or both false.

Same type. Their actual type is not determined.

Mutual accusations

A says “B is a knave.”

B says “A is a knave.”

Using the same variables,

$$S_A = \neg q, \quad S_B = \neg p.$$

Therefore,

$$(p \leftrightarrow \neg q) \wedge (q \leftrightarrow \neg p) \equiv (p \wedge \neg q) \vee (\neg p \wedge q) \equiv p \oplus q.$$

Opposite types.

Which person is the knight is not determined.

Knights and knaves practice

Use the same p , q , S_A , and S_B .

A says

"B is a knave."

B says

"We are opposite types."

1. Identify S_A and S_B .
2. Write the two speaker constraints.
3. Determine the types of A and B.

Summary

1. Propositions have truth values. Connectives build formulas.
 2. Truth tables define meaning and test equivalence.
 3. Implications express requirements.
 4. Solve by cases or by equivalence laws.
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Next lecture

Predicate logic: adding objects, properties, and quantifiers.

Equivalence laws: basic reference

Law	Equivalent formulas
identity	$p \wedge T \equiv p; p \vee F \equiv p$
domination	$p \vee T \equiv T; p \wedge F \equiv F$
idempotent	$p \vee p \equiv p; p \wedge p \equiv p$
double negation	$\neg\neg p \equiv p$
complement	$p \vee \neg p \equiv T; p \wedge \neg p \equiv F$
commutative	$p \vee q \equiv q \vee p; p \wedge q \equiv q \wedge p$
absorption	$p \vee (p \wedge q) \equiv p; p \wedge (p \vee q) \equiv p$

Apply any law in either direction.

Equivalence laws: structural reference

Law	Equivalent formulas
associative	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
distributive	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Conditional equivalences: reference

Type	Equivalent formulas
conditional	$p \rightarrow q \equiv \neg p \vee q$
	$p \rightarrow q \equiv \neg q \rightarrow \neg p$
	$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
	$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
	$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
	$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$
biconditional	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
	$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
	$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$